

THE SPECIOUS FAMILY

Why some mixers only look the part.

A new group of cheap hash finalizers, found by enumeration. Every member mapped, every one refuted at depth. A find-and-autopsy report. The family is not a usable mixer and is not recommended here. Its value is the map and the mechanism its death confirms.

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What this is about

The cheapest known 64-bit hash finalizer that passes the RRC ladder costs 9 operation points and has one particular shape: a constant onto the input, then two 128-bit folds (the wyhash pattern). The question of this report: **at cost 9, is that shape the only one?**

Answering it meant enumerating instead of searching further. The space of three-op chains of cost class 9 over a pool of proven constants holds 30,980 members in three shape families. Enumeration starts where a search stops: it does not decide in advance which corner of a space is worth looking at.

How far it actually got is stated here rather than implied, because the difference decides what may be concluded from it. The run of 19 August 2026 worked through family A (fold $\rightarrow$ x $\rightarrow$ fold) and decided **1,309 chains** before a deliberate cap stopped it: too many candidates had reached the expensive 2 GB stage (90 against a limit of 80) to continue without looking at the situation first. Families B and C were never started. This is therefore an enumeration in method and a fragment in coverage: about a quarter of one family out of three.

That is enough to prove such a shape *exists*, which is what this report is about. It is not enough to prove it is the only other one. An enumeration that runs to the end cannot miss anything; this one did not run to the end, and the unopened remainder is the honest open question at the back of this report.

The result is a double one, and both halves are the finding: there *is* a second shape that passes the 2 GB entry test — a whole family of it. And it dies as a body the moment you measure deeper.

The find

The family has the shape **fold** → **byte swap** → **fold**:

```
v = mulfold(v, c1); v = bswap(v); v = mulfold(v, c2);
```

Cost: $4 + 1 + 4 = 9$. The byte swap (bswap) is a single x86 instruction that reverses the eight bytes of a word. **23 constant combinations** of this shape pass the ladder's entry test: 32 counter-stream disguises at 2 GB each, not one may fall. Before this enumeration, only the wyhash pattern could do that at cost 9.

What that entry test could not see.

Its 32 streams are four transformations against eight of the 64 rotations, so seven eighths of the rotation space never runs. The depth data of this very family shows what that costs: **23 stream deaths land below 2 GB** — the entry test's own volume — spread over 9 of the 23 chains, and every one of them sits on a rotation the entry test never runs. Not one on a rotation it does. These chains did not survive the entry test because 2 GB was too little; they survived it because the disguise that kills them was not among the eight. Counted from the depth result file, all 23 candidates, 1,472 streams.

Why the search never found them.

The published comparison tables of recent years do not contain it either: anyone measuring mixers against RRC streams usually tests twenty hand-picked constructions, and a shape nobody writes down never gets tested. An enumeration closes that gap.

An evolutionary niche search over the same space (467 chains, 60 generations) brushed the family exactly once: one member at 30 of 32 streams. Just short, archived as a niche, never grown into a family. The enumeration had 19 of them after 71 minutes and 23 after 116, inside a quarter of a single family. Where a space is finite, walking it beats guessing at it, even when the walk gets cut short.

How high the bar sits that they later fail.

For scale: the industry classics Stafford mix13 and the Murmur3 finalizer (both cost 12) die on the same streams at 64–128 KB. Every member of this family holds 2 to 16 GB — the *rejects* of this report outlive the state of the art by orders of magnitude, at a quarter less cost. (In fairness: the classics were never designed against RRC streams.)

The map: all 23 members

Constants in short form; the key is below the table. “Clean” = streams with no finding in the depth test (64 streams at 16 GB each), followed by the earliest failure. Fourteen of the 23 first fail at 2³¹ or later; nine fail earlier, the first of them at 2¹⁹.

c1	c2	Clean	Earliest fail
bf5847...	94d049...	51/64	2 ²⁹
bf5847...	3c79ac...	55/64	2 ³²
bf5847...	1c69b3...	53/64	2 ³¹
94d049...	94d049... (c1 = c2!)	54/64	2 ³¹
94d049...	3c79ac...	54/64	2 ³¹
94d049...	1c69b3...	54/64	2 ³¹
94d049...	c4ceb9...	54/64	2 ³¹
3c79ac...	781f94...	53/64	2 ³¹
9e6c63...	94d049...	52/64	2 ³¹
9e6d62...	94d049...	54/64	2 ²⁹
c4ceb9...	94d049...	52/64	2 ³¹
c4ceb9...	3c79ac...	55/64	2 ³²
c4ceb9...	1c69b3...	55/64	2 ³²
c4ceb9...	9e6d62...	51/64	2 ¹⁹
c4ceb9...	781f94...	55/64	2 ³²
d6e8fe...	3c79ac...	53/64	2 ³²
d6e8fe...	1c69b3...	54/64	2 ³¹
9fb21c...	94d049...	49/64	2 ²⁵
9fb21c...	3c79ac...	52/64	2 ²⁸
9fb21c...	9e6d62...	49/64	2 ²⁹
9fb21c...	c4ceb9...	48/64	2 ²⁸
9fb21c...	781f94...	51/64	2 ³⁰
a24bae...	94d049...	54/64	2 ³⁰

Short forms: bf5847... = 0xbf58476d1ce4e5b9, 94d049... = 0x94d049bb133111eb (both Stafford mix13) · 3c79ac... = 0x3c79ac492ba7b653, 1c69b3... = 0x1c69b3f74ac4ae35 (Moremur, Evensen) · c4ceb9... = 0xc4ceb9fe1a85ec53 (Murmur3, Appleby) · 9e6c63... = 0x9e6c63d0676a9a99, 9e6d62... = 0x9e6d62d06f6a9a9b (NASAM, Evensen) · d6e8fe... = 0xd6e8feb86659fd93, 9fb21c... = 0x9fb21c651e98df25, a24bae... = 0xa24baed4963ee407 (constant pool) · 781f94... = 0x781f94b96e8edb3b (mulfold champion, own search).

Observations.

First: the choice of constants barely matters — nine different c_1 values are alive, and even $c_1 = c_2$ passes the entry test. The active ingredient is the byte swap, not the constant. Second: mix13's second constant is the most frequent partner, seven times as the second constant, and present in ten of the 23 chains altogether. Constants of a mixer that itself dies between 64 KB and 2 MB carry up to 16 GB here. Third: the 9fb21c group is consistently the weakest (48–52 clean), and the family has inner structure that would be worth a finer mechanism analysis.

The autopsy

The depth test runs 64 streams at 16 GB (2^{34}), the same protocol the wyhash pattern passes 64/64. **All 23 members** failed: 48 to 55 clean streams, failures between 2^{31} and 2^{34} (a few earlier). Not one member reached the next stage.

The signature in the death pattern.

Of 260 individual failures, **260 lie on streams without complement** ($217 \times T0C0$, $43 \times T1C0$, zero on $C=1$). That is the known weakness of the bare double fold: inputs of low multiplicative weight (small counters, many zero bits) produce a thin 128-bit product, and the fold has nothing to fold. The complement makes inputs generic, which is why that half is spotless.

What the byte swap actually does.

The byte swap between the folds tears up the byte alignment of the thin product — far enough that the test program no longer sees the structure at 2 GB. But it does not make the input generic: given eight times the data, PractRand finds the structure again. **The byte swap disguises the weakness; the constant up front removes it.** That is where this family differs from the equally priced wyhash pattern, and why one holds 2^{34} and the other does not.

With that, the mechanism from the cost-limit paper has its second, unexpected confirmation: at this price the cost point that closes the K8 weakness has to make the *input* generic. A permutation of the *intermediate product* for the same money looks identical in the entry test, and is not.

What one more point buys in the same position.

That reading holds only at nine. It would be wrong to conclude that the middle position is the mistake, and the counter-example comes from this rig: **meer10** carries the very same skeleton (fold, something, fold), repairs in the middle like this family does, and passes the diploma, further than the input-side repair has ever reached (2^{40} against lu9's 2^{38}). The difference is one cost point, and what it is spent on:

Middle step	Cost	What it does	Result
bswap (this family)	9	rearranges bytes, adds no diffusion	dies 2^{31} – 2^{34}
$x \wedge= (x \gg 28) \wedge (x \gg 6)$	10	diffuses — carries move down, not only up	passes 2^{40}

A byte swap moves bits; a double xor-shift mixes them. Same position, one point apart, and the one hides the weakness while the other heals it. That is what gives this family its lasting value: it is a graveyard of 23 chains, and it is also the **control group** that shows what the missing point actually buys. It explains, mechanically, how little separates the ladder's entry test from its diploma: one member of a three-operation skeleton.

Three lessons

For building mixers.

Between two equally priced shapes that the entry test cannot tell apart, what decides is where the repair step acts: before the first fold (heals) or between the folds (disguises). Anyone designing finalizers should put the +1 step on the input.

For measurement methodology.

An entry test can be fooled by more than individual outliers: an entire, systematically constructible family will do it. Depth staging is therefore not a gesture of thoroughness but a duty: without the 2^{34} stage, 23 false champions would be standing here.

For searching.

Where the space is finite, exhaustion is preferable to heuristics: it finds reliably, and it can also say “nothing” reliably.

What this is — and what it is NOT

It IS: a fully mapped candidate family.

The path to the find, all 23 members, death depths and death patterns are documented and reproducible; every run carried a positive and a negative control (mix13 must fail, the wyhash pattern must hold).

It is NOT a usable mixer.

Nobody should deploy a member of this family. It passes the entry test and dies at depth. Its value is epistemic: the map and the confirmed mechanism.

Novelty: the shape is known, its measurement is not.

The literature check of 20 August found the shape itself twice: Remco Bloemen's *MulSwapHash* (2014) is exactly mul-bswap-mul with plain 64-bit multiplications, tested only with SMHasher; and Scala's standard library ships a mul-reverseBytes-mul mixer as *byteswap32/64* (Phil Bagwell) — without any published quality characterisation. Deployed hashes also use the chain in partial paths (CityHash64, XXH3-128). What is NEW in this report is therefore not the shape but two things: the 128-bit fold variant, and the first RRC/PractRand characterisation at all: entry passed, depth refuted, mechanism named. For the relatives in Scala and CityHash (plain multiplications instead of folds) our finding is a testable hypothesis, not a verdict. The constants are deliberately borrowed; our own constants demonstrably fail.

The measured depth is named.

Measurement went to 2^{34} per stream. No member reached the diploma stage (RRC-64-40, 1 TB per stream): after 0 of 23 at depth it was no longer the question.

Reproducibility

Every member is a three-op chain over documented constants and runs through the ladder's generic chain feeder without recompiling. Enumeration driver, depth-test driver and test protocol are documented; the verdict depends only on the byte stream, not on the machine.

In brief

The find	A second shape passes the K9 entry test: fold → byte swap → fold, 23 constant combinations, found in 1,309 enumerated chains of a 30,980-chain space, before a cap stopped the run.
The autopsy	0 of 23 survive the depth test (2^{34}); all 260 failures on streams without complement, the signature of the fold weakness.
The mechanism	bswap disguises the weakness (enough for 2 GB), the constant up front removes it (holds 2^{34}). The repair step has to make the input generic.
The status	A fully mapped candidate family, not a product. Novelty subject to the literature check. No member reached the diploma stage.
The value	The map, the confirmed mechanism and the evidence that entry tests can be fooled family-wide. Depth staging is a duty.

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As of: 20 August 2026 (working draft). Measuring rig: see *The Ladder*. Cost frame and K9 lower bound: see *The Cost Limit*. The chain that does hold the depth this family fails, one cost class higher: see *MEER10*.